JEE Physics

Detailed JEE Main + Advanced Syllabus

Exam Coverage	JEE Main + JEE Advanced aligned topic map
Format	Module-wise syllabus with detailed bullet-wise subtopics
Use Case	Foundation planning, course brochure, chapter scheduling and curriculum design

Structured Syllabus

Unit 1: Units, Measurements and Mathematical Tools

- System of units, dimensions, dimensional analysis, significant figures, error analysis and graphical interpretation of experimental data.
- Scalars and vectors, vector algebra, unit vectors, resolution of vectors and applications in mechanics.

Unit 2: Kinematics

- Motion in one dimension and two dimensions; displacement, velocity, acceleration and relative motion.
- Projectile motion, uniform circular motion and graphical interpretation of kinematical equations.

Unit 3: Laws of Motion and Friction

- Newton's laws of motion, free-body diagrams, pseudo force, equilibrium and constrained motion.
- Static and kinetic friction, angle of friction, angle of repose and connected-body problems.

Unit 4: Work, Energy, Power and Centre of Mass

- Work-energy theorem, conservative and non-conservative forces, power and conservation of mechanical energy.
- System of particles, centre of mass motion, impulse, momentum conservation and collision problems.

Unit 5: Rotational Motion and Gravitation

- Moment of inertia, radius of gyration, torque, rolling motion, angular momentum and rotational equilibrium.
- Universal law of gravitation, gravitational potential, escape velocity, satellites and Kepler's laws.

Unit 6: Properties of Matter and Fluid Mechanics

- Elasticity, stress-strain relation, Young's modulus and energy stored in stretched wire.
- Surface tension, viscosity, streamline flow, Bernoulli's theorem and applications such as Venturimeter and lift.

Unit 7: Thermal Physics

- Thermal expansion, calorimetry, kinetic theory of gases, ideal gas equation and RMS speed.
- First law of thermodynamics, isothermal/adiabatic processes, heat engines, refrigerators and entropy-level conceptual understanding.

Unit 8: Oscillations and Waves

- Simple harmonic motion, spring-block systems, pendulum, energy in SHM, damping, resonance and forced oscillation.

- Wave motion, superposition, standing waves, Doppler effect, beats and organ pipes.

Unit 9: Electrostatics and Capacitance

- Coulomb’s law, electric field, field lines, electric flux and Gauss’s law with standard symmetric charge distributions.
- Electric potential, equipotential surfaces, potential energy, capacitors, dielectric and combinations of capacitors.

Unit 10: Current Electricity and Heating Effect

- Current density, drift velocity, resistivity, temperature dependence of resistance and Ohm’s law.
- Kirchhoff’s laws, Wheatstone bridge, potentiometer, electrical power and heating effect of current.

Unit 11: Magnetism, EMI and AC

- Magnetic field due to current, Biot–Savart law, Ampere’s law, Lorentz force and motion of charged particle in fields.
- Electromagnetic induction, Lenz’s law, self and mutual inductance, alternating current, impedance, resonance and transformers.

Unit 12: Optics and Modern Physics

- Ray optics: mirrors, lenses, lens-maker formula, prism, optical instruments and defects of vision.
- Wave optics: interference, Young’s double slit experiment, diffraction and polarization.
- Photoelectric effect, Bohr model, nuclei, radioactivity, semiconductors, logic gates and communication basics.